

Recall our basic probabilistic experiment of tossing a biased coin n times.

This is a very simple model, yet

surprisingly powerful. Many important 概率 distributions that are widely used to model 实数-world

phenomena can be derived from looking at this basic coin-tossing model.

The first example is the Binomial (n, p) 分布, introduced earlier.

This is the distribution of the number

of Heads, S , in n tosses of a biased coin with 概率 p to be Heads. 回忆 the 分布 of S

n

is $P[S = k] = \binom{n}{k} p^k (1-p)^{n-k}$ 对于 $k \in \{0, 1, \dots, n\}$. The expected value is $E[S] = np$ 和 the 方差 is

$np(1-p)$

$\text{Var}(S) = np(1-p)$. The 二项的 分布 is frequently used to model the number of successes in a

n

repeated experiment.

1 Geometric 分布

Consider repeatedly tossing a biased coin with Heads probability p .

Let X denote the number of tosses until

the first Head appears. 那么 X is a random 变量 that takes values in $\mathbb{Z}^+$, the 集合 of 正的 integers.

The event that X

$= i$ is equal to the event of observing Tails for the first $i-1$ tosses and getting Heads in the

i th toss, which occurs with 概率 $(1-p)^{i-1}p$. Such a random 变量 is

called a geometric random

变量. 笔记 that X is unbounded: i.e., 对于 any 正的 整数 i (no matter how large), 存在 some

positive probability that $X = i$.

The geometric distribution frequently occurs in applications because we are often interested in how long we have to wait before a certain event happens:

how many runs before the system fails, how many shots before

one is on target, how many polls samples before we find a Democrat, how many retransmissions of a packet before successfully reaching the destination, etc.

Definition 18.1 (Geometric Distribution). A random variable X

for which

$P[X = i] = (1-p)^{i-1}p$, for $i = 1, 2, 3, \dots$,

is said to have the geometric distribution with parameter p .

This is abbreviated as $X \sim \text{Geometric}(p)$ 或

$X \sim \text{Geom}(p)$.

As a quick check, we can verify that the total probability of X

is equal to 1:

$\sum_{i=1}^{\infty} (1-p)^{i-1}p = p \sum_{i=1}^{\infty} (1-p)^{i-1} = p \times \frac{1}{1-(1-p)} = 1$,

wherein the second-to-last step we have used the formula for a geometric series.

If we plot the distribution of X

(i.e., the values $P[X=i]$ against i) we get a curve that decreases monotonically by a factor of $1-p$ at each step, as illustrated in Figure 1.

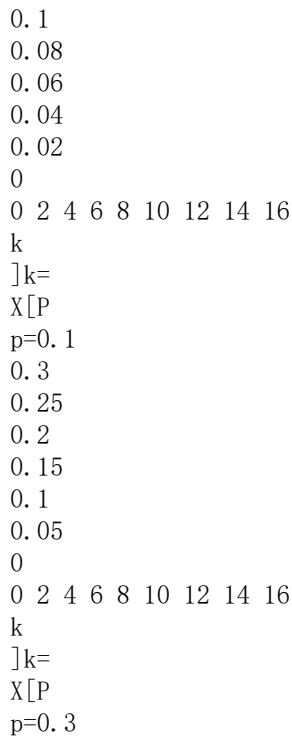


Figure1: Illustration of the Geometric(p) distribution for $p=0.1$ and $p=0.3$.

1.1 Mean 和 方差 of a Geometric Random 变量

Let us now compute the expectation $E[X]$.

Applying the definition of expected value directly gives us:

$$E[X] = \sum_{i=1}^{\infty} i \times P[X=i] = p \sum_{i=1}^{\infty} i(1-p)^{i-1}.$$

然而, the final summation is a little tricky to evaluate (though it can be done). Instead, we will use the

following alternative formula for expectation that simplifies the calculation and is useful in its own right.

Theorem 18.1 (Tail Sum Formula). Let X

be a random variable that takes values in $\{0, 1, 2, \dots\}$. 那么

$$E[X] = \sum_{i=1}^{\infty} P[X \geq i].$$

证明. 对于 notational convenience, 令's write $p = P[X=i]$, 对于 $i = 0, 1, 2, \dots$. From the 定义 of

期望, we have

$$\begin{aligned} E[X] &= (0 \times p) + (1 \times p) + (2 \times p) + (3 \times p) + (4 \times p) + \dots \\ &= p + (p + p) + (p + p + p) + (p + p + p + p) + \dots \\ &= (p + p + p + \dots) + (p + p + p + \dots) + (p + p + \dots) + (p + \dots) + \dots \\ &= P[X \geq 1] + P[X \geq 2] + P[X \geq 3] + P[X \geq 4] + \dots \end{aligned}$$

In the third line, we have regrouped the terms into convenient infinitesimal sums, and each infinitesimal sum is exactly the probability that $X \geq i$ for each i .

You should check that you understand how the fourth line follows from the third.

Let us repeat the proof more formally, this time using more compact mathematical notation:

$$E[X] = \sum_{j=1}^{\infty} j \times P[X = j] = \sum_{j=1}^{\infty} \sum_{i=1}^j P[X = j] = \sum_{i=1}^{\infty} \sum_{j=i}^{\infty} P[X = j] = \sum_{i=1}^{\infty} P[X \geq i],$$

where the third equality follows from interchanging the order of summations.
CS70, Spring 2026, Note 18.2

We can now use Theorem 18.1 to compute $E[X]$ more easily.

Theorem 18.2. For $X \sim \text{Geometric}(p)$, we have $E[X] = 1/p$.

证明. The key observation is that for a geometric random variable X ,
 $P[X \geq i] = (1-p)^{i-1}$ for $i = 1, 2, \dots$ (1)

We can obtain this simply by summing $P[X = j]$ 对于 $j \geq i$.

Another way of seeing this is to 笔记 that the

事件 " $X \geq i$ " means at least i tosses are required. This is 等价

to saying that the first $i-1$ tosses are

all Tails, and the probability of this event is precisely $(1-p)^{i-1}$.

Now, plugging (1) into Theorem 18.1, we

get

$$E[X] = \sum_{i=1}^{\infty} P[X \geq i] = \sum_{i=1}^{\infty} (1-p)^{i-1} = \frac{1}{1-(1-p)} = \frac{1}{p},$$

where we have used the formula for geometric series.

所以, the expected number of tosses of a biased coin until the first Head appears is

1. Thus for a fair coin, the

expected number of tosses 直到 the first Head is 2 (但是 of course the actual number of tosses can be any positive integer).

Let us now compute the variance of X .

Theorem 18.3. For $X \sim \text{Geometric}(p)$, we have $\text{Var}(X) = 1/p^2$.

证明.

We use the formula

$$\text{Var}(X) = E[X^2] - (E[X])^2 = E[X^2] - 1/p^2$$

$$= \sum_{i=1}^{\infty} i^2 (1-p)^{i-1} p$$

Thus we just have to compute

$$E[X^2] = \sum_{i=1}^{\infty} i^2 (1-p)^{i-1} p$$

, which we can do using a calculus trick as follows. Starting from

$$\sum_{i=0}^{\infty} (1-p)^i = \frac{1}{1-(1-p)} = \frac{1}{p}$$

and differentiating with respect to p , we get

$$\sum_{i=0}^{\infty} i (1-p)^{i-1} (-1) = -\frac{1}{p^2}$$

Herewe dropped the first term, for $i=0$, since its derivative is zero, and multiplied both sides by -1 .

(笔记

incidentally that (3) gives an alternative proof of the fact that $E[X] = 1/p$; why?)

Now we multiply both sides of (3) by $(1-p)$ and differentiate again to get

$$\sum_{i=0}^{\infty} i^2 (1-p)^{i-1} (-1) = \frac{2}{p^3}$$

$$\sum_{i=1}^3 (1-p)^{i-1} p = 1$$
 Since the left-hand side here is exactly $E(X)$, we conclude that $E(X) = \frac{1}{1-p}$.
 Finally, plugging this into (2) gives us $Var(X) = \frac{p}{(1-p)^2}$, as claimed.
 CS70, Spring 2026, Note 18.3

1.2 Memoryless Property

The geometric 分布 has a special property called memorylessness. To get some intuition 对于 this

property, suppose we revisit our earlier motivational example of repeatedly tossing a biased coin with heads 概率 p , and let X denote the number of tosses until the first head appears. 因此, $X \sim \text{Geometric}(p)$.

As we have seen, the probability that we need more than n tosses before getting the first head is $P[X > n] = (1-p)^n$, since the first n tosses must all have been tails.

What 如果 我们有 already tossed the coin m times without getting a head? What is the 概率 that we need more than n additional tosses before getting our first head?

We can calculate this probability as:

$$\begin{aligned}
 P[X > n+m] &= P[X > n+m | X > m] \\
 &= P[X > m] \\
 &= (1-p)^{n+m} \\
 &= (1-p)^m (1-p)^n \\
 &= (1-p)^n P[X > n].
 \end{aligned}$$

This gives us the formal statement of the memoryless property of geometric distributions:

$$P[X > n+m | X > m] = P[X > n].$$

The geometric distribution is memoryless in the sense that the length of time we have already been waiting does 非 affect the additional time 我们有 to wait 直到 we get our first head. (Indeed, 如果 我们有 tossed a

fair coin n times and it has come up tails every time, the probability that it comes up heads on the $(n+1)$ th toss is still $1/2$.)

1.3 Application: Coupon Collecting

Recall the coupon collecting problem from a previous note: we are trying to collect a set of n different base-

ball cards by buying boxes of cereal, each of which contains exactly one random card (uniformly distributed among the n cards).

How many boxes do we need to buy until we have collected at least one copy of every

card?

令 S 表示 the number of boxes we need to buy in order to collect all n cards. The 分布 of S is

n n

difficult to 计算 directly (try it 对于 $n=3$), 和 we did some calculations in a previous 笔记 to estimate the likely values of S . 但是 如果 we are only interested in the expected value $E[S]$, 那么 we can evaluate it

n n

easily using linearity of expectation and what we have just learned about the geometric distribution.

We start by writing

$$S = X_1 + X_2 + \dots + X_n \quad (4)$$

n 1 2 n

对于 simpler random variables X_i , 其中 X_i is the number of boxes we buy 当...时 trying to get the i th new

i i

card (starting immediately after we have gotten the $(i-1)$ st new card).

With this definition, make sure you

believe (4) before proceeding.

What does the distribution of X_i look like? Well, X_i is trivial:

no matter what happens, we always get a new

i 1

card in the first box (since we have none to start with). So $P[X_i = 1] = 1$, and thus $E[X_i] = 1$.

1 1

This property is shared by the exponential distribution, the continuous analog of the geometric distribution, which we'll cover

in a future note. Indeed, these are the only two distributions that are memoryless.

CS70, Spring 2026, Note 18.4

How about X_2 ? Each time we buy a box, we will get the same old card with 概率 $1/n$, 和 a new

2 n

card with probability $1/n$.

So we can think of buying boxes as flipping a biased coin with Heads probability

n

$p = 1/n$; then X_2

is just the number of tosses until the first Head appears. So X_2 has the geometric distribution

n 2 2

with parameter $p = 1/n$, 和

n

n

$E[X_2] = \frac{1}{p} = n$.

2

n 1

How about X_3 ? This is very similar to X_2

except that now we only get a new card with probability $1/n^2$ (因为

3 2 n

there are now two old ones). So X_3

has the geometric distribution with parameter $p = 1/n^2$, 和

3 n

n

$E[X_3] = \frac{1}{p} = n^2$.

3

n 2

Arguing in the same way, we see that, 对于 $i=1, 2, \dots, n$, X_i has the geometric 分布 with parameter

$$p = \frac{1}{n} \sum_{i=1}^n i, \text{ and hence that } E[X] = \frac{1}{n} \sum_{i=1}^n i.$$
 Finally, applying linearity of expectation to (4), we get

$$E[S_n] = \sum_{i=1}^n E[X_i] = \sum_{i=1}^n \frac{1}{n} \sum_{j=1}^n j = \frac{1}{n} \sum_{j=1}^n j \cdot n = \sum_{j=1}^n j. \quad (5)$$

This is an exact expression for $E[S_n]$. We can obtain a tidier form by noting that the sum in (5) actually has

a very good approximation, 2 即:

$$\sum_{i=1}^n i \approx \ln n + \gamma,$$

where $\gamma = 0.5772\dots$ is known as Euler's constant.

因此, the expected number of cereal boxes needed to collect n cards is about $n(\ln n + \gamma)$. This is an excel-

lent 近似 to the exact formula from (5) even 对于 quite small values of n . 所以 对于 例子, 对于 $n=100$, we expect to buy about 518 boxes.

2 泊松 分布

考虑 the number of clicks of a Geiger counter, 其中 measures radioactive emissions. The average

number of such clicks per unit time, λ , is a measure of radioactivity, but the actual number of clicks fluctuates according to a certain 分布 called the 泊松 分布. What is remarkable is that the average

value, λ , completely determines the probability distribution on the number of clicks X .

Definition 18.2 (Poisson distribution). A random variable X for which

$$P[X = i] = \frac{e^{-\lambda} \lambda^i}{i!}, \text{ for } i = 0, 1, 2, \dots \quad (6)$$

is said to have the Poisson distribution with parameter λ . This is abbreviated as $X \sim \text{Poisson}(\lambda)$.

To make sure this is a valid definition, let us check that (6) is in fact a distribution, i.e., that the probabilities sum to 1. We have

$$\sum_{i=0}^{\infty} \frac{e^{-\lambda} \lambda^i}{i!} = e^{-\lambda} \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{-\lambda} e^{\lambda} = 1.$$

2 This is another of the little tricks you might like to carry around in your toolbox. CS70, Spring 2026, Note 18.5

In the second-to-last step, we used the Taylor series expansion $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

+ • • • .

2! 3!

The 泊松 分布 is a very widely accepted model 对于 所以-called “rare events,” such as disconnected

phonecalls, radioactiveemissions, crossoversinchromosomes, thenumberofcasesofdisease, thenumber of births per hour, etc. This model is appropriate whenever the occurrences can be assumed to happen randomlywithsomeconstantdensity λ inacontinuousregion (oftimeorspace), suchthatoccurrencesin disjoint subregions are 独立的. One can 那么 证明 that the number of occurrences in a region of unit sizesouldobeythePoissondistributionwithparameter λ .

例子

Supposewhenwewriteanarticle, wemakeanaverageofltypoperpage.

We canmodelthiswithaPoisson

randomvariableX with $\lambda = 1$.

Sotheprobabilitythatapagehas5typosis

$15 \cdot 1 \cdot 1$

$P[X = 5] = e^{-1} \approx .$

$5! \cdot 120e^{-326}$

Now 假设 the article has 200 pages. 如果 we 假设 the numbers of typos on each page are 独立的,

thentheprobabilitythatthereisatleastonepagewithexactly5typosis

$P[\text{apagewithexactly5typos}] = 1 - P[\text{everypagehas } \neq 5 \text{ typos}]$

200

$= 1 - \prod_{k=1}^{200} P[\text{pagekhas } \neq 5 \text{ typos}]$

$k=1$

200

$= 1 - \prod_{k=1}^{200} (1 - P[\text{pagekhasexactly5typos}])$

$k=1$

$(\text{cid:18}) (\text{cid:19}) 200$

1

$= 1 - 1 \approx 0.46,$

$120e$

whereinthelaststepwehaveusedourearliercalculationfor $P[X = 5]$.

2.1 Mean 和 方差 of a 泊松 Random 变量

LetusnowcalculatetheexpectationandvarianceofaPoissonrandomvariable.

Theorem18.4. ForaPoissonrandomvariableX 泊松 (λ), wehave $E[X] = \lambda$

and $\text{Var}(X) = \lambda$.

证明.

We cancalculate $E[X]$ directly from the definition of expectation:

∞

$E[X] = \sum_{i=0}^{\infty} i \times P[X = i]$

$i=0$

$\infty \lambda i$

$= \sum_{i=0}^{\infty} i \times e^{-\lambda} \frac{\lambda^i}{i!}$ (the $i=0$ term is equal to 0 so we omit it)

$i!$

$i=1$

$\infty \lambda i \cdot 1$

$= \lambda e^{-\lambda} \sum_{i=1}^{\infty} (i-1)!$

$(i-1)!$

$i=1$

$\infty \lambda j$

$= \lambda e^{-\lambda} \sum_{j=0}^{\infty} \lambda$ (since $\lambda = \sum_{j=0}^{\infty} 1$ with $j=i-1$)

$j!$

j=0
= λ .
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Similarly, we can calculate $E(\text{cid}:2) \times 2(\text{cid}:3)$ as follows:

$$\begin{aligned}
& \infty \\
& E(\text{cid:2}) \quad X2(\text{cid:3}) = \sum i2 \times P[X = i] \\
& i=0 \\
& \infty \quad \lambda \quad i \\
& = \sum i2 \quad e^{-\lambda} \quad (\text{the } i=0 \text{ term is equal to } 0 \text{ so we omit it}) \\
& i! \\
& i=1 \\
& \infty \quad \lambda \quad i \quad 1 \\
& = \lambda e^{-\lambda} \sum i \\
& (i-1)! \\
& i=1 \\
& (\text{cid:34}) \quad (\text{cid:35}) \\
& \infty \quad \lambda \quad i \quad 1 \quad \infty \quad \lambda \quad i \quad 1 \\
& = \lambda e^{-\lambda} \sum + \sum (\text{replacing } i \text{ by } (i-1)+1) \\
& (i-2)! \quad (i-1)! \\
& i=2 \quad i=1 \\
& (\text{cid:104}) \quad (\text{cid:105}) \quad \infty \quad \lambda \quad j \\
& = \lambda e^{-\lambda} \quad \lambda e^{-\lambda} + e^{-\lambda} \quad (\text{using } e^{-\lambda} = \sum \text{ twice, with } j=i-1 \text{ and } j=i-2) \\
& j! \\
& j=0 \\
& = \lambda^2 + \lambda.
\end{aligned}$$

因此,

$$\text{Var}(X) = E(X^2) - E(X)^2 = \lambda^2 + \lambda - \lambda^2 = \lambda,$$

as desired.

A plot of the Poisson distribution reveals a curve that rises monotonically to a single peak and then decreases monotonically. The peak is as close as possible to the expected value, i.e., at $i = \lambda$. 图 2 illustrates the $\text{Poisson}(\lambda)$ distribution for $\lambda = 1, 2, 5, 20$.

2.2 Sum of 独立的 泊松 Random Variables

As illustrated in Figure 2, the $\text{Poisson}(\lambda)$ distribution becomes more symmetric and resembles a “bell curve” as λ increases. As we will learn later, this phenomenon is closely related to the following useful fact regarding a sum of independent Poisson random variables.

Theorem 18.5. Let X be Poisson(λ) and Y

泊松(μ)be independent Poisson random variables. 那么,
 $X+Y$ 泊松(λ+μ).

证明. For all $k=0, 1, 2, \dots$, we have

$$\begin{aligned}
& P[X+Y=k] = \sum_{j=0}^k P[X=j, Y=k-j] \\
& = \sum_{j=0}^k P[X=j]P[Y=k-j] \\
& = \sum_{j=0}^k e^{-\lambda} \frac{\lambda^j}{j!} e^{-\mu} \frac{\mu^{k-j}}{(k-j)!} \\
& = e^{-(\lambda+\mu)} \sum_{j=0}^k \frac{\lambda^j \mu^{k-j}}{j!(k-j)!} \\
& = e^{-(\lambda+\mu)} \frac{(\lambda+\mu)^k}{k!}
\end{aligned}$$

$$= e^{-(\lambda + \mu)} \frac{(\lambda + \mu)^k}{k!},$$

where these two inequalities follow from independence, and the last inequality from the binomial theorem. CS70, Spring 2026, Note 18.7

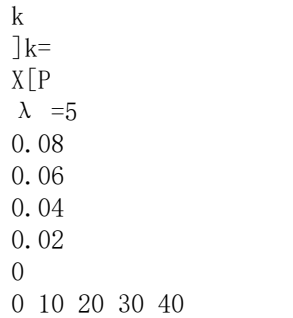
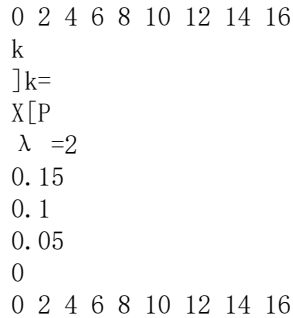
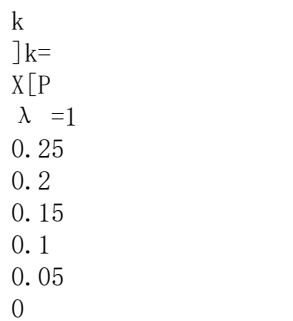


Figure 2: Illustration of the Poisson(λ) distribution for $\lambda = 1, 2, 5, 20$.

By induction, we conclude that if $X_1, X_2, \dots, X_n$ are independent Poisson random variables with parameters

$\lambda_1, \lambda_2, \dots, \lambda_n$, respectively, then

$X_1 + X_2 + \dots + X_n$ is Poisson($\lambda_1 + \lambda_2 + \dots + \lambda_n$).

2.3 泊松 as 极限 of 二项的

To see a concrete example of how the Poisson distribution arises, suppose we want to model the number of cell phone users initiating calls in a network during a time period, of duration (say) 1 minute. There

are many customers in the network, 和 all of them can potentially make a call during this time period. 然而, only a very small fraction of them actually will. Under this scenario, it seems reasonable to make two assumptions:

- The probability of having more than 1 customer initiating a call in any small time interval is negligible.
-

The initiation of calls in disjoint time intervals are independent events. 那么, 如果 we divide the one-minute time period into n disjoint intervals, the number of calls X in that time period can be modeled as a 二项的 (n, p) random 变量, 其中 p is the 概率 of having a call

initiated in a time interval of length $1/n$. 但是 what should p be in terms of the relevant parameters of the 题目? 如果 calls are initiated at an average rate of λ calls per minute, 那么 $E[X] = \lambda$ 和 所以 $np = \lambda$, i.e., CS70, Spring 2026, Note 18.8

$p = \lambda/n$. So $X \sim \text{二项的}(n, \lambda/n)$.

As we shall see below, as we let $n \rightarrow \infty$, this distribution tends

to the Poisson distribution with parameter λ .

This explains why the Poisson distribution is a model for “rare events”: it counts the number of heads in a very long 序列 ($n \rightarrow \infty$) of coin flips, 其中 the expected number of heads is a small finite number.

Now we will prove the claim above that $\text{Poisson}(\lambda)$ is the limit of $\text{Binomial}(n, \lambda/n)$, as $n \rightarrow \infty$.

Theorem 18.6. Let $X \sim \text{二项的}(n, \lambda/n)$ where $\lambda > 0$ is a fixed constant. Then for every $i = 0, 1, 2, \dots$,

$$P[X = i] \rightarrow \frac{e^{-\lambda} \lambda^i}{i!} \text{ as } n \rightarrow \infty.$$

That is, the probability distribution of X converges to the Poisson distribution with parameter λ .

证明. Fix $i \in \{0, 1, 2, \dots\}$, 和 假设 $n \geq i$ (because we will let $n \rightarrow \infty$). 那么, because X has 二项的 distribution with parameter n and $p = \lambda/n$,

$$P[X = i] = \binom{n}{i} p^i (1-p)^{n-i}$$

$$= \frac{n!}{i!(n-i)!} \left(\frac{\lambda}{n}\right)^i \left(1 - \frac{\lambda}{n}\right)^{n-i}$$

$$= \frac{n!}{i!(n-i)!} \frac{\lambda^i}{n^i} \left(1 - \frac{\lambda}{n}\right)^{n-i}$$

$$= \frac{n!}{i!(n-i)!} \frac{\lambda^i}{n^i} \left(1 - \frac{\lambda}{n}\right)^{n-i}$$

$$= \frac{n!}{i!(n-i)!} \frac{\lambda^i}{n^i} \left(1 - \frac{\lambda}{n}\right)^{n-i}$$

$$P[X = i] = \frac{n!}{i!(n-i)!} \frac{\lambda^i}{n^i} \left(1 - \frac{\lambda}{n}\right)^{n-i}$$

Let us collect the factors into

$$\frac{\lambda^i}{i!} \left(1 - \frac{\lambda}{n}\right)^{n-i}$$

$$\frac{n!}{(n-i)!} \left(1 - \frac{\lambda}{n}\right)^{n-i}$$

$$= \frac{\lambda^i}{i!} \left(1 - \frac{\lambda}{n}\right)^{n-i}$$

$$\frac{n!}{(n-i)!} \left(1 - \frac{\lambda}{n}\right)^{n-i}$$

$$= \frac{\lambda^i}{i!} \left(1 - \frac{\lambda}{n}\right)^{n-i}$$

$$\frac{n!}{(n-i)!} \left(1 - \frac{\lambda}{n}\right)^{n-i}$$

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